

Notes: Topel and Ward (QJE, 1992)

Job Mobility and the Careers of Young Men

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Longitudinal Employee-Employer Data	4
3.2	Sample construction	4
3.3	Key variables	5
3.4	Main descriptive data patterns	5
4	Models and empirical specifications	5
4.1	Within-job wage growth	5
4.2	Between-job wage gains	6
4.3	Job-shopping mobility model	6
4.4	Proportional hazards estimation	6
5	Identification and empirical design	7
5.1	What the design can identify	7
5.2	Key identification logic	7
5.3	Selection and heterogeneity concerns	7
5.4	Limitations	7
6	Tables and figures	8
6.1	Table I: Age Distributions at Onset of Continuous Work and Table II: Distribution of Actual Labor Market Experience	8
6.2	Table III: Distribution of Cumulative Jobs and Figure I: Exit Hazard by Current Job Tenure	8
6.3	Table IV: Mobility from New Jobs by Experience and Prior Jobs and Table V: Mobility from Current Job by Job Experience	9
6.4	Table VI: Determinants of Within-Job Wage Growth and Table VII: Wage Changes at Job Transitions	10
6.5	Figure II: Mobility by Experience and Figure III: Mobility by Tenure	11
6.6	Table VIII: Proportional Hazards Models and Table IX: Initial Wages and Wage Growth	12
7	Short summary	13
8	Conclusion	14
8.1	Core conclusion	14
8.2	Economic intuition	14
8.3	Incremental contribution revisited	14
8.4	Implications for labor economics	14
8.5	Limitations	14

8.6	Takeaway	15
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1 Big picture

- **Question.** How do young workers move from fragile early employment relationships to stable adult careers, and how much of early-career wage growth is generated by job mobility?
- **Setting.** The paper studies young white men in the United States using the Longitudinal Employee-Employer Data (LEED), an administrative Social Security data set that records quarterly earnings by employer from 1957 to 1972.
- **Main empirical fact 1: early careers involve very high mobility.** A typical worker holds around seven full-time jobs during the first ten years in the labor market. This is roughly two thirds of his lifetime job changes.
- **Main empirical fact 2: early careers also involve rapid wage growth.** Wage growth is concentrated early in the career, and wage gains at job changes account for at least one third of wage growth in the first ten years.
- **Main empirical fact 3: job mobility declines sharply with tenure and experience.** Many jobs end almost immediately: more than one third of new jobs end within three months and about two thirds end within one year.
- **Interpretation.** Young workers engage in job shopping. They learn about jobs and about their own comparative advantage by sampling employers. Successful matches survive; poor matches end quickly.
- **Central mechanism.** The job-specific wage is a key state variable. Holding fixed individual heterogeneity, a higher wage makes a worker substantially less likely to leave a job.
- **Economic takeaway.** Early-career turnover is not merely wasteful instability. It is a sorting mechanism that helps workers find better and more durable employment relationships.

2 Incremental contribution of the paper

- **From descriptive turnover to career dynamics.** Earlier work documented that mobility declines with age and tenure. This paper connects the mobility process to wage growth and career development.
- **Administrative employer-linked evidence.** The paper uses quarterly employer-specific Social Security earnings records rather than self-reported annual survey histories. This lets the authors observe short jobs, quick transitions, and employer changes with much more precision.
- **A clean view from labor-market entry.** Because the sample follows workers from the beginning of their careers, the paper avoids the usual left-censoring problem in job-duration studies.
- **Quantification of mobility's contribution to wage growth.** The paper shows that between-job wage changes are not a minor side issue: they account for roughly one third or more of early-career wage growth.
- **A direct test of job-shopping models.** By conditioning mobility hazards on current job-specific wages, the paper tests whether wages act as sufficient statistics for match quality.

- **Reversal of the experience effect.** Unconditionally, job mobility declines with experience. Conditional on the wage, the effect reverses: more experienced workers are more likely to leave a job at the same wage, consistent with rising outside options.
- **Bridge between human capital and search.** The paper challenges the standard view that early-career wage growth is mainly due to general human capital accumulation. Search and matching are central parts of the wage-growth process.

3 Data and measurement

3.1 Longitudinal Employee-Employer Data

- The LEED file contains quarterly employee-employer records for more than one million individuals.
- For each worker-employer-quarter, the data report Social Security earnings credited to the worker, the number of employees in the firm, detailed industry, and employer location.
- Each worker and each employer has a unique identifier, allowing the authors to identify job starts, job endings, returns to previous employers, and job-to-job transitions.
- The panel begins in 1957 and ends in 1972.
- Personal characteristics are limited: the data contain age, race, and sex, but not schooling, hours worked, or weeks worked.

Interpretation: The strength of the data is precise employer-linked quarterly earnings histories. The weakness is that quarterly earnings are treated as a wage proxy because hours and weeks are not observed.

3.2 Sample construction

- The authors focus on young white males whose careers can be observed from labor-market entry.
- Jobs are measured from the first job held on or after the worker's eighteenth birthday.
- A quarter is classified as full-time participation if total earnings across all employers exceed 70 percent of the minimum quarterly wage, assuming full-time work.
- The major employer in a quarter is the employer contributing the most to total earnings.
- Job tenure is measured as continuous quarters with a single major employer.
- Temporary one-quarter changes in the identity of the major employer are smoothed over, so they do not mechanically create job changes.
- Records with continuous nonemployment gaps of two years or more are excluded to reduce ambiguity about uncovered employment or military service.

Interpretation: The data construction is designed to identify stable employer relationships while not mistaking short reporting or temporary-layoff patterns for true job separations.

3.3 Key variables

- **Potential experience** is time since first entry to the labor market.
- **Actual experience** is the cumulative amount of full-time work time.
- **Job tenure** is continuous time with the current employer.
- **Job changes** are changes in the identity of the major employer after smoothing over short transitory changes.
- **Wages** are proxied by real quarterly earnings in full-time quarters, excluding first and last quarters of jobs when necessary because earnings may cover only part of a quarter.

3.4 Main descriptive data patterns

- New entrants have weak labor-force attachment: early employment spells often end in nonemployment.
- Workers gradually move toward full-time, stable attachment over the first several years.
- New jobs are extremely fragile, but the job exit hazard declines sharply with job tenure.
- Workers accumulate many jobs early: by ten years after entry, the average worker has held roughly seven full-time jobs.

4 Models and empirical specifications

4.1 Within-job wage growth

The paper starts from a standard earnings equation with experience and current job tenure:

$$w_{jt} = H(X_{jt}, T_{jt}) + \phi_j + \varepsilon_{jt}, \quad (1)$$

where w_{jt} is log quarterly earnings on job j at time t , X_{jt} is labor-market experience, T_{jt} is current tenure, ϕ_j is a job-specific fixed component, and ε_{jt} is a time-varying earnings component.

Differencing within jobs removes the job fixed effect:

$$\Delta w_{jt} = \Delta H(X_{jt}, T_{jt}) + \Delta \varepsilon_{jt}. \quad (2)$$

- This specification estimates predictable wage growth within jobs as a function of experience and tenure.
- It also lets the authors examine the autocovariance structure of within-job wage innovations.
- A key result is that within-job earnings evolve approximately as a random walk with drift.

Interpretation: If within-job earnings follow a random walk with drift, the current wage summarizes the value of the job for future wage prospects. Past wage changes add little predictive information.

4.2 Between-job wage gains

For a transition from job $j - 1$ to job j , the authors want to estimate the wage change at the transition. Since first and last job quarters may be partial quarters, they use earnings in the second quarter of the new job and the penultimate quarter of the old job:

$$\begin{aligned} E(w_{jt} - w_{j-1,t-1} \mid w_{j,t+1}, w_{j-1,t-2}) &= w_{j,t+1} - w_{j-1,t-2} \\ &\quad - E(w_{j,t+1} - w_{jt} \mid \cdot) \\ &\quad - E(w_{j-1,t-1} - w_{j-1,t-2} \mid \cdot). \end{aligned} \tag{3}$$

The last two terms adjust for predicted within-job growth around the transition.

Interpretation: This method isolates wage gains associated with changing employers rather than confusing job-change gains with normal within-job wage growth.

4.3 Job-shopping mobility model

The empirical model is motivated by wealth-maximizing on-the-job search. External wage offers satisfy

$$\Pr(w^o < z; X) = G(z; X), \quad G_X(z; X) < 0, \tag{4}$$

so outside wage offers improve with experience.

Within-job wage offers depend on the current wage, experience, and tenure:

$$\Pr(w^i < y; w, X, T) = F(y; w, X, T). \tag{5}$$

The value of remaining with a current job is $V(w, X, T)$. A job change occurs when the value of an outside offer exceeds the value of the current job. This defines a reservation wage $R(w, X, T)$, and the hazard of leaving the current job is

$$\lambda(w, X, T) = \pi [1 - G(R(w, X, T); X)], \tag{6}$$

where π is the offer-arrival rate.

Main predictions:

- A higher current wage lowers mobility because it raises the value of the current match.
- Conditional on wage and tenure, experience may increase mobility because more experienced workers have better outside offers.
- The unconditional decline of mobility with experience can therefore reverse once the current wage is controlled for.

4.4 Proportional hazards estimation

The empirical hazard model is

$$\lambda(T; Z_{ij}(T), \mu_i) = \exp\{h(T) + Z_{ij}(T)' \beta + \mu_i\}, \tag{7}$$

where $h(T)$ captures tenure dependence, $Z_{ij}(T)$ includes the wage, experience, tenure, firm size, season, and transition controls, and μ_i is a worker fixed effect.

- The worker fixed effect captures time-invariant individual heterogeneity in mobility, such as permanent mover-stayer tendencies.
- The model is estimated on workers with at least thirteen years of potential post-entry labor-market experience.
- The main estimates use 872 workers and 44,089 job quarters.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies empirical relationships between job mobility, wages, experience, and tenure using within-person career histories.
- The hazard models identify how changes in wages within a worker's career predict the probability of leaving the current job.
- The design is strongest for distinguishing job-shopping models from purely descriptive turnover models because the wage provides an observed measure of match quality.

5.2 Key identification logic

- Unconditionally, mobility declines with experience and tenure.
- In a search model, this decline can reflect sorting into higher-wage, better matches.
- Therefore, if the current wage is an important state variable, controlling for the wage should change the estimated role of experience and tenure.
- The paper finds exactly this for experience: unconditional mobility declines with experience, but wage-conditioned mobility rises with experience.
- This reversal supports the view that workers with more experience have better outside opportunities, and that observed stability partly reflects successful sorting into high-wage matches.

5.3 Selection and heterogeneity concerns

- Workers differ in their baseline tendency to move. Some may be persistent movers because of preferences, occupations, industries, or unmeasured traits.
- Jobs differ in match quality. High-quality matches should have higher wages and longer duration.
- The paper addresses individual heterogeneity by estimating fixed-effect proportional hazard models.
- It addresses job-specific heterogeneity by conditioning on current job wages and wage growth.

5.4 Limitations

- Quarterly earnings are not pure hourly wages because hours and weeks are unavailable.
- Schooling is not observed, so the sample cannot directly separate high school graduates from college entrants.
- The data cover young men entering the labor market in the late 1950s and 1960s; external validity to later labor markets, women, and college-intensive careers is not automatic.
- Some nonemployment spells may actually be uncovered employment, military service, or part-time work.

6 Tables and figures

6.1 Table I: Age Distributions at Onset of Continuous Work and Table II: Distribution of Actual Labor Market Experience

Read:

- Table I shows that many young men begin some employment very early, but longer employment spells start later. For example, short spells are common by age 18, whereas spells lasting at least one year are much more concentrated at later ages.
- Table II shows that only 16.2 percent of workers with ten years of potential experience are continuously employed for all forty quarters.
- Table II also shows that actual market experience accumulates slowly early on: during the first potential year, workers accumulate only 0.70 years of actual experience.
- By later years, actual experience accumulates much more quickly, with additional experience close to 0.90 years per potential year by the ninth and tenth years.

Economic message: Entry into the labor market is gradual rather than instantaneous. The early career involves weak labor-force attachment, short employment spells, and nonemployment transitions before workers settle into steadier work.

TABLE I
AGE DISTRIBUTIONS AT ONSET OF CONTINUOUS WORK—SIXTEEN YEARS
OF AGE AND OLDER ($N = 8,102$)

Length of spell	Age at beginning of employment spell							
	≤ 18	19	20	21	22	23	24	≥ 25
≥ 1 quarter	46.0	25.6	14.7	7.6	4.1	1.5	0.5	0.1
≥ 2 quarters	29.6	24.9	18.8	11.4	8.1	4.8	1.7	0.7
≥ 3 quarters	24.2	22.7	18.6	12.9	10.1	7.0	2.8	1.5
≥ 1 year	21.6	21.0	17.3	13.6	11.8	8.3	3.9	2.6

Note. Figures in the table refer to the percentage of all workers who begin an employment (not job) spell of the indicated length at the indicated age. For this table only we consider jobs that begin as early as age sixteen.

TABLE II

A. Distribution of actual labor market experience for persons with ten years of potential experience (eighteen years of age and older)

	Years					Quarters								
	≤ 4	5	6	7	8	33	34	35	36	37	38	39	40	
Percent with indicated experience	0.8	1.7	4.6	8.4	15.8	6.2	5.9	6.0	6.7	8.2	8.8	10.8	16.2	

B. Actual labor market experience by years of potential experience

	Potential market experience (years)									
	1	2	3	4	5	6	7	8	9	10
Actual market experience	0.70	1.36	2.10	2.89	3.73	4.61	5.49	6.38	7.29	8.19
Additional experience	0.70	0.66	0.74	0.79	0.84	0.88	0.88	0.89	0.91	0.90

Table I: Age distributions at onset of continuous work

Table II: Actual and potential labor-market experience

6.2 Table III: Distribution of Cumulative Jobs and Figure I: Exit Hazard by Current Job Tenure

Read:

- Table III shows that cumulative full-time jobs rise very rapidly in the first ten years. The average worker has 4.61 jobs by six years of potential experience and 6.96 jobs by ten years.
- More than half of workers have held at least six jobs by ten years of potential experience.
- The annual flow of new jobs declines as experience accumulates: the first year of actual experience is split among many jobs, while later years involve fewer new jobs.
- Figure I plots the empirical quarterly exit hazard by current tenure. The hazard is extremely high at the start of a job and then falls sharply.
- More than one third of new jobs end within three months, and about two thirds end within one year.

Economic message: Early-career employment is a high-frequency matching process. The steep fall in the exit hazard with tenure suggests that surviving jobs are better matches or involve stronger attachment.

TABLE III

A. Distribution of cumulative full-time jobs for persons with 10 years of potential experience

	Cumulative full-time jobs										
	1	2	3	4	5	6	7	8	9	10	11-15
Percent with indicated number of jobs	4.3	7.0	9.9	11.1	11.6	9.5	9.0	8.2	6.9	4.8	13.0

B. Average Cumulative full-time jobs by years of labor market experience

	Years of experience									
	1	2	3	4	5	6	7	8	9	10
By potential experience	1.60	2.47	3.24	3.92	4.56	5.14	5.66	6.09	6.54	6.96
New jobs	1.60	0.87	0.77	0.68	0.64	0.58	0.52	0.43	0.45	0.42
By actual experience	2.78	3.77	4.50	4.99	5.37	5.62	5.83	6.11	6.18	6.22
New jobs	2.78	0.99	0.73	0.49	0.38	0.25	0.21	0.28	0.07	0.04

Note. Potential experience is time since entry, actual experience is time employed. "Jobs" refers to the number of new employers during the indicated time span. Multiple employers at any date are not counted.

Table III: Cumulative full-time jobs

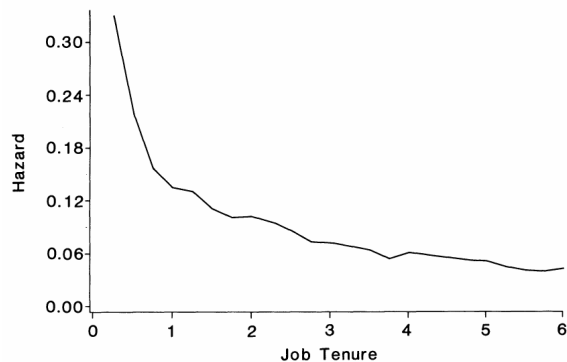


FIGURE I
EXIT HAZARD BY CURRENT JOB TENURE

6.3 Table IV: Mobility from New Jobs by Experience and Prior Jobs and Table V: Mobility from Current Job by Job Experience

Read:

- Table IV Panel A shows that new jobs become more stable as workers gain prior full-time experience, but turnover remains substantial even after many years of work.
- Table IV Panel B shows mobility by number of prior jobs. Workers with more prior jobs tend to be more likely to leave a new job, consistent with individual heterogeneity in mobility propensities.
- Table V decomposes job endings into transitions to new jobs and transitions to nonemployment.
- For the first job, a large fraction of early endings result in nonemployment rather than immediate job-to-job transitions.
- As experience accumulates, the relative importance of job-to-job transitions rises and the importance of exits to nonemployment declines.

Economic message: Early mobility reflects both weak labor-force attachment and active job shopping. The transition from unstable early careers to stable employment involves fewer exits to nonemployment and more sorting among jobs.

TABLE IV
EMPIRICAL MOBILITY FUNCTIONS BY PRIOR MARKET EXPERIENCE,
NUMBER OF PRIOR JOBS

	Current job tenure (quarters)												
	1	2	3	4	5	6	7	8	9	10	11	12	13
A. Prior experience:													
None	0.39	0.20	0.15	0.15	0.17	0.13	0.10	0.12	0.12	0.09	0.08	0.08	0.08
1 year or less	0.41	0.25	0.17	0.14	0.16	0.11	0.11	0.11	0.11	0.09	0.08	0.07	0.07
1-2 years	0.34	0.24	0.17	0.13	0.13	0.13	0.11	0.11	0.09	0.09	0.08	0.08	0.07
2-4 years	0.30	0.23	0.16	0.13	0.12	0.11	0.10	0.10	0.09	0.08	0.07	0.06	0.07
4-8 years	0.27	0.20	0.15	0.13	0.10	0.10	0.09	0.09	0.08	0.08	0.05	0.06	0.05
More than 8 years	0.22	0.19	0.14	0.12	0.10	0.09	0.08	0.06	0.05	0.05	0.05	0.06	0.05
B. Prior jobs:													
None	0.39	0.20	0.15	0.15	0.17	0.13	0.10	0.12	0.12	0.09	0.08	0.08	0.08
1	0.34	0.21	0.16	0.13	0.13	0.11	0.10	0.10	0.10	0.10	0.08	0.07	0.07
2	0.32	0.21	0.14	0.12	0.12	0.10	0.10	0.10	0.09	0.08	0.07	0.07	0.07
3 or 4	0.29	0.21	0.15	0.12	0.12	0.10	0.09	0.09	0.09	0.09	0.08	0.07	0.07
5-7	0.30	0.22	0.17	0.13	0.11	0.12	0.10	0.10	0.09	0.08	0.06	0.07	0.06
8 or more	0.37	0.30	0.22	0.18	0.15	0.14	0.14	0.11	0.09	0.12	0.05	0.06	0.08
C. Aggregate hazard	0.33	0.22	0.16	0.14	0.13	0.11	0.10	0.10	0.10	0.08	0.07	0.07	0.07

Note. Figures refer to the percentage of the relevant group that changes major employer between successive quarters. $N = 58,181$.

TABLE V
MOBILITY FROM CURRENT JOB BY TYPE OF TRANSITION AND EXPERIENCE AT START OF JOB

		Current job tenure (quarters)												
		1	2	3	4	5	6	7	8	9	10	11	12	13
Prior experience:														
None	<i>j-j</i>	0.093	0.078	0.059	0.070	0.068	0.066	0.054	0.072	0.065	0.050	0.047	0.053	0.045
	<i>j-n</i>	0.297	0.118	0.091	0.080	0.099	0.059	0.050	0.047	0.054	0.035	0.036	0.023	0.032
1 year or less	<i>j-j</i>	0.150	0.122	0.094	0.083	0.083	0.064	0.078	0.075	0.068	0.076	0.058	0.055	0.047
	<i>j-n</i>	0.262	0.125	0.073	0.062	0.075	0.050	0.036	0.039	0.045	0.034	0.028	0.028	0.029
1-2 years	<i>j-j</i>	0.183	0.144	0.109	0.089	0.084	0.079	0.083	0.078	0.067	0.063	0.056	0.050	0.050
	<i>j-n</i>	0.156	0.094	0.060	0.040	0.044	0.048	0.025	0.031	0.026	0.023	0.020	0.033	0.018
2-4 years	<i>j-j</i>	0.186	0.151	0.120	0.096	0.086	0.078	0.074	0.070	0.067	0.060	0.054	0.048	0.049
	<i>j-n</i>	0.109	0.075	0.043	0.036	0.036	0.029	0.025	0.027	0.025	0.021	0.018	0.017	0.021
4-7 years	<i>j-j</i>	0.193	0.145	0.114	0.098	0.080	0.067	0.071	0.069	0.054	0.060	0.039	0.048	0.041
	<i>j-n</i>	0.078	0.057	0.035	0.029	0.121	0.029	0.020	0.017	0.022	0.018	0.015	0.015	0.012
More than 8 years	<i>j-j</i>	0.175	0.140	0.109	0.092	0.070	0.075	0.065	0.049	0.042	0.044	0.041	0.048	0.043
	<i>j-n</i>	0.048	0.049	0.032	0.025	0.026	0.019	0.011	0.014	0.008	0.008	0.013	0.016	0.011

Note. Estimates are the percentage moving within one quarter for the indicated reason. *j-j* represents job-to-job transitions, *j-n* represents job-to-"nonemployment" transitions. See text for definitions. Sample is 9,919 white males.

Table IV: Mobility by prior experience and prior jobs

Table V: Types of job endings

6.4 Table VI: Determinants of Within-Job Wage Growth and Table VII: Wage Changes at Job Transitions

Read:

- Table VI shows that within-job wage growth is high early in the career and declines with experience.
- The average annual wage growth within jobs is about 7 percent in the wage-growth sample, and around 14 percent for new entrants.
- The residual autocovariance evidence implies that within-job wage innovations are close to a random walk with drift.
- Table VII shows that job transitions generate large wage gains: the average job transition in the first ten years is associated with a wage increase of about 12 percent, and the adjusted gain is about 10 percent.
- Job-transition gains are largest early in the career and decline with experience.
- The cumulative gain from job transitions accounts for roughly one third to 40 percent of early-career wage growth, depending on the accounting measure.

Economic message: Wage growth is not only within-job human capital accumulation. Employer changes are a major component of wage growth and are closely tied to the process of finding better matches.

TABLE VI
DETERMINANTS OF WITHIN-JOB WAGE GROWTH

A. Wage growth models ($\Delta W = 0.071$)						
	Intercept	ΔX^2	ΔT^2	ΔT^3	Completed tenure	Job change
(i)	0.136 (0.006)	-0.005 (0.0004)	—	—	—	—
(ii)	0.138 (0.006)	-0.004 (0.0004)	-0.002 (0.0005)	—	—	—
(iii)	0.151 (0.007)	-0.004 (0.0004)	-0.006 (0.0015)	0.0003 (0.0001)	—	—
(iv)	0.129 (0.009)	-0.004 (0.0005)	-0.009 (0.0017)	0.0003 (0.0001)	0.004 (0.0010)	—
(v)	0.139 (0.0104)	-0.004 (0.0005)	-0.008 (0.0017)	0.0004 (0.0001)	0.003 (0.0012)	-0.013 (0.007)
Mean	13.29	6.66	56.15	7.00	0.28	
(S.d.)	(6.67)	(5.49)	(85.93)	(3.67)	(0.45)	
B. Residual autocorrelations for $\Delta \epsilon_{jt}$: model (v)						
Lag	0	1	2	3	4	
Covariance	0.0434 (0.0020)	-0.0132 (0.0014)	-0.0002 (0.0009)	-0.0003 (0.0008)	-0.0007 (0.0009)	
Autocorrelation	1.00	-0.3269	-0.0043	-0.0088	-0.0173	
C. Estimated error structure (i)						
	ρ	$\sigma_{\epsilon v}$	$\sigma_{\eta \eta}$			

Table VI: Within-job wage growth

TABLE VII
WAGE CHANGES AT JOB TRANSITIONS FOR YOUNG MEN

A. Average wage changes at job transitions as a component of wage growth: experience interval (years)					
	0-2.5	2.5-5	5-7.5	7.5-10	10-15
Average wage change at job transitions	0.171 (0.015)	0.119 (0.016)	0.079 (0.015)	0.057 (0.016)	0.114 (0.007)
Average wage gain at job transitions	0.145 (0.015)	0.099 (0.016)	0.064 (0.015)	0.046 (0.016)	0.094 (0.007)
Cumulative wage change at transitions	0.169 (0.014)	0.118 (0.015)	0.054 (0.015)	0.039 (0.015)	0.380 (0.022)
Cumulative wage gains at transitions	0.143 (0.013)	0.091 (0.015)	0.039 (0.015)	0.032 (0.015)	0.313 (0.021)
Cumulative wage growth	0.316 (0.012)	0.286 (0.013)	0.204 (0.012)	0.141 (0.013)	0.947 (0.015)
B. Determinants of wage changes at job transitions					
Dependent variable	Intercept	ΔX^2	T_{j-1}	T_{j-1}^2	Δ (completed tenure)
$Y_{jt} - Y_{j,t-1}$	0.1992 (0.0168)	-0.0013 (0.0003)	-0.0257 (0.0042)	0.0016 (0.0003)	-2.4 ⁻⁵ (5.4 ⁻⁶)
$Y_{jt} - Y_{j,t-2}$	0.2305 (0.0151)	-0.0015 (0.0003)	-0.0272 (0.0042)	0.0016 (0.0030)	-2.5 ⁻⁵ (5.4 ⁻⁶)
$Y_{jt} - Y_{j,t-3}$	0.1684 (0.0168)	-0.0010 (0.0003)	-0.0249 (0.0042)	0.0016 (0.0003)	-2.5 ⁻⁵ (5.4 ⁻⁶)
$Y_{jt} - Y_{j,t-4}$	0.1998 (0.0151)	-0.0013 (0.0003)	-0.0264 (0.0042)	0.0016 (0.0003)	-2.5 ⁻⁵ (5.4 ⁻⁶)

Notes: Standard errors are in parentheses. Dependent variable is the change in log quarterly earnings at job transitions. The estimates are calculated from job transitions that result in a new job lasting more than one quarter. There were 1,987 such transitions in the data. See text for description of procedures. In panel B job tenure and experience are measured in quarters.

Table VII: Wage changes at job transitions

6.5 Figure II: Mobility by Experience and Figure III: Mobility by Tenure

Read:

- Figure II compares the estimated probability of changing jobs by labor-market experience with and without controlling for the current wage.
- Without wage controls, mobility declines with experience.
- With wage controls, the relationship reverses: holding the wage constant, more experienced workers are more likely to change jobs.
- Figure III shows that mobility declines with current job tenure, and this pattern is not materially changed by conditioning on the wage.

Economic message: Figure II is one of the most important pieces of evidence. The wage control reveals that experienced workers have better outside options; the unconditional decline in mobility reflects sorting into better-paying matches. Figure III indicates that tenure-related stability is not fully explained by the current wage.



FIGURE II
Estimated Probability of Changing Jobs by Years of Labor Market Experience

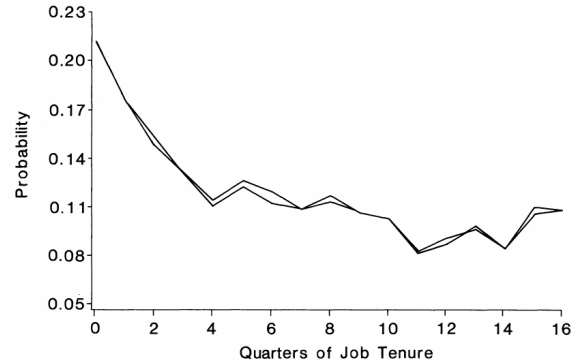


Figure III: Mobility by current job tenure

6.6 Table VIII: Proportional Hazards Models and Table IX: Initial Wages and Wage Growth

Read:

- Table VIII estimates proportional hazard models for job mobility with worker fixed effects.
- The current wage has a large negative effect on the probability of leaving a job. In the preferred specifications, the coefficient on current wage is around -1.6 to -1.8 .
- The authors interpret this as implying that a 10 percent wage gain reduces the quarterly probability of changing jobs by about two percentage points at sample means.
- Once the current wage is controlled for, the prior-job indicator becomes small and statistically weak, suggesting that wage captures much of the value of a job.
- Small-firm jobs have much higher turnover than large-firm jobs, even after controlling for wages and fixed worker differences.
- Table IX shows that the initial wage on a job also strongly predicts lower mobility when it is the only wage control.
- When both current and initial wages are included, current wage dominates, but the initial wage still has some independent role.
- Jobs with higher past wage growth are less likely to end, conditional on current wage.

Economic message: The formal hazard models confirm the job-shopping interpretation: high-wage matches survive. Wage levels and wage growth are central to the movement from unstable early jobs to stable employment relationships.

TABLE VIII
ESTIMATED PROPORTIONAL HAZARDS MODELS FOR JOB MOBILITY, LEED WHITE
MALES WITH THIRTEEN+ YEARS OF EXPERIENCE

	Mean (s.d.)	(1)	(2)	(3)	(4)
1. Current wage	2.780 (0.472)	—	-1.700 (0.059)	-1.637 (0.060)	-1.77 (0.059)
× quarter 1	0.338 (0.877)	—	-0.236 (0.048)	-0.211 (0.052)	-0.303 (0.052)
× quarter 2	0.254 (0.785)	—	-0.093 (0.032)	-0.125 (0.033)	-0.128 (0.033)
2. Prior job	0.623 (0.484)	-0.221 (0.040)	-0.012 (0.042)	0.042 (0.043)	0.032 (0.043)
3. Experience spline:					
a. 0–1 year	2.88 (0.506)	0.045 (0.028)	0.081 (0.029)	0.038 (0.036)	0.035 (0.036)
b. 1–2 years	6.847 (2.507)	-0.016 (0.007)	0.053 (0.008)	0.051 (0.013)	0.055 (0.013)
c. >2 years	15.989	-0.014	0.013	0.022	0.023
4. Tenure spline:					
a. 0–1 year	2.33 (1.105)	-0.189 (0.016)	-0.343 (0.047)	-0.448 (0.072)	-0.437 (0.072)
b. 1–2 years	3.892 (3.571)	-0.039 (0.008)	-0.028 (0.008)	-0.092 (0.020)	-0.104 (0.020)
c. >2 years	3.915 (7.601)	0.016 (0.005)	0.014 (0.005)	0.015 (0.008)	0.015 (0.008)
5. Experience × tenure interac- tions?	—	No	No	Yes	Yes
6. Firm size:					
a. 1–9	0.063 (0.243)	0.863 (0.095)	0.566 (0.096)	0.571 (0.097)	—
b. 10–99	0.321 (0.467)	0.670 (0.091)	0.473 (0.091)	0.475 (0.092)	—
c. 100–499	0.232 (0.422)	0.442 (0.096)	0.331 (0.096)	0.333 (0.097)	—
d. 500–999	0.055 (0.227)	0.099 (0.116)	0.053 (0.116)	0.056 (0.116)	—
e. 1000–2499	0.066 (0.249)	-0.186 (0.126)	-0.231 (0.126)	-0.223 (0.123)	—
f. Not reported	0.212 (0.409)	0.978 (0.092)	0.534 (0.093)	0.533 (0.093)	—
7. Season					
Winter	0.240 (0.427)	0.267 (0.041)	0.285 (0.041)	0.288 (0.041)	0.292 (0.041)
Spring	0.244 (0.430)	0.170 (0.039)	0.172 (0.039)	0.173 (0.039)	0.174 (0.039)
Fall	0.260 (0.439)	0.272 (0.040)	0.292 (0.040)	0.293 (0.040)	0.297 (0.040)
8. Individual effects					
Mean	—	-2.206	-2.227	-2.328	-2.957
(s.d.)	—	(1.238)	(0.825)	(0.845)	(0.868)
9. Log likelihood	—	-12,951	-12,351	-12,321	-12,370
10. Observations	44,089				

Note. Figures in parentheses are asymptotic standard errors. Estimates for the tenure spline are evaluated at twelve years of experience. Six tenure-experience interactions are not reported. Not reported are 872 individual effects.

TABLE IX
ESTIMATED HAZARD FUNCTIONS FOR JOB MOBILITY:
INITIAL WAGES AND WAGE GROWTH

	1	2	3
1. Initial wage	-1.475 (0.061)	0.473 (0.150)	—
2. Wage growth	—	—	-5.318 (1.276)
3. Current wage	—	-2.09 (0.143)	-1.616 (0.061)
× quarter 1	-0.387 (0.051)	-0.330 (0.052)	-0.331 (0.052)
× quarter 2	-0.185 (0.033)	-0.147 (0.034)	-0.148 (0.034)
4. Prior job	0.032 (0.043)	0.034 (0.043)	0.035 (0.043)
5. Experience spline:			
a. 0–1 year	0.032 (0.036)	0.038 (0.036)	0.038 (0.036)
b. 1–2 years	0.047 (0.013)	0.050 (0.013)	0.050 (0.013)
c. >2 years	0.019 (0.003)	0.021 (0.003)	0.021 (0.003)
6. Tenure spline:			
a. 0–1 year	-0.541 (0.072)	-0.476 (0.072)	-0.483 (0.072)
b. 1–4 years	-0.094 (0.021)	-0.091 (0.020)	-0.087 (0.020)
c. >4 years	-0.009 (0.008)	0.024 (0.009)	0.018 (0.008)
7. Experience × tenure interactions?	Yes	Yes	Yes
8. Individual effects			
Mean	-2.099	-2.247	-2.249
(s.d.)	(0.859)	(0.845)	(0.845)
9. Log likelihood	-12,418	-12,311	-12,307
10. Observations	44,089		

Note. Figures in parentheses are asymptotic standard errors. Estimates for the tenure spline are evaluated at the mean of experience. Other regressors from Table VIII are included in the model, but are not reported separately. Not reported are 872 individuals.

Table VIII: Estimated proportional hazards models

Table IX: Initial wages and wage growth

7 Short summary

- Topel and Ward study job mobility and wage growth among young men using quarterly employer-linked Social Security records.
- The first ten years of work contain most lifetime job changes and a large share of lifetime wage growth.
- New jobs are very fragile: more than one third end within a quarter, and about two thirds end within a year.
- A typical young worker holds about seven jobs in the first ten years after entry.
- Wage gains at job changes are large, especially early in the career, and account for at least one third of early-career wage growth.
- Within-job wages evolve approximately as a random walk with drift, making current wages a key statistic for match quality.

- Conditional on current wages, the effect of experience on mobility reverses: experienced workers are more likely to leave the same-wage job because they have better outside options.
- Fixed-effect hazard models show that higher current wages sharply reduce job-exit probabilities.
- The paper concludes that early-career job mobility is a productive search and sorting process, not simply instability.

8 Conclusion

8.1 Core conclusion

Topel and Ward show that job mobility is central to the early careers of young men. Rather than treating job changes as disruptions to human capital accumulation, the paper argues that job changes are a key mechanism through which workers discover better matches and raise their wages. The movement toward stable employment relationships occurs through a period of intensive job sampling.

8.2 Economic intuition

Young workers begin their careers with limited information about employers, jobs, and their own comparative advantage. They sample jobs, observe wages and match quality, leave low-wage or poor-growth jobs, and stay in better-paying matches. As a result, turnover is high early in the career but falls as workers sort into higher-wage jobs. Wage growth and job stability are two sides of the same matching process.

8.3 Incremental contribution revisited

The paper's contribution is not only to document that young workers move a lot. Its deeper contribution is to show that mobility is linked to wage growth in a way predicted by job-shopping and matching models. The use of administrative employer-linked data lets the authors observe many short jobs and transitions that survey data often miss. This changes the interpretation of early-career earnings growth: search and match formation are as important as standard human capital accumulation.

8.4 Implications for labor economics

The results imply that models of lifetime earnings should treat early-career job search as a productive investment activity. Policies or institutions that restrict mobility may slow the process by which young workers find high-quality matches. At the same time, early-career instability may be less harmful than it appears if mobility leads to higher wages and more durable employment.

8.5 Limitations

The paper studies young white men entering the U.S. labor market in an earlier period, so the findings may differ for women, minorities, college-educated workers, and modern labor markets with different institutions. The data do not measure hours, schooling, job tasks, or voluntary versus involuntary separations. Quarterly earnings are an imperfect wage measure, although the administrative nature of the data reduces reporting error relative to surveys.

8.6 Takeaway

The enduring takeaway is that early-career job mobility is a sorting technology. Young workers change jobs frequently because changing jobs is how many of them find better matches, earn large wage gains, and eventually settle into stable employment relationships.